

Finding and Fixing Related-Rates and Optimization Setup Errors

- Most problems below show a student's work. The arithmetic is usually fine — the *setup* is where the mistake lives.
- When a part asks you to name or describe an error, write a sentence. A circled line with no explanation earns no credit.
- Keep exact values (fractions, π) unless a decimal is requested.

1. Air is pumped into a spherical balloon so that its radius grows at the constant rate $\frac{dr}{dt} = 3$ cm/s. Using $V = \frac{4}{3}\pi r^3$, find $\frac{dV}{dt}$ at the instant when $r = 5$ cm. Round to two decimal places (cubic centimetres per second).

Answer: _____

2. A cube's edge length s grows at $\frac{ds}{dt} = 0.5$ cm/s. Asked for $\frac{dV}{dt}$ when $s = 4$ cm, a student writes:

$$V = s^3 \quad \implies \quad \frac{dV}{dt} = 3s^2 = 3(4)^2 = 48.$$

(a) Name the error in the student's differentiation.

Answer: _____

(b) Give the corrected value of $\frac{dV}{dt}$.

Answer: _____

3. A rectangular pen is fenced against a straight barn wall, so only three sides need fence and 60 m of fence is available. A student writes: *"Let x be each side perpendicular to the barn, so $2x + y = 60$ and $A = xy = x(60 - 2x)$. Setting $A = 0$ gives $x = 0$ or $x = 30$, so the answer is $x = 30$ m."*

(a) Explain what went wrong in the student's last step.

Answer: _____

(b) Find the corrected value of x that maximizes the area.

Answer: _____

4. A 13 ft ladder leans against a vertical wall. Its foot slides away from the wall at $\frac{dx}{dt} = 2$ ft/s. Asked for $\frac{dy}{dt}$ when the foot is $x = 5$ ft from the wall (so $y = 12$ ft), a student writes: “From $x^2 + y^2 = 169$ I get $2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$, so $\frac{dy}{dt} = \frac{5}{6}$ ft/s.”

(a) Describe the error in the student’s final line.

Answer: _____

(b) Give the corrected value of $\frac{dy}{dt}$, in feet per second.

Answer: _____

5. An open-top box is folded from a 20 cm by 20 cm sheet by cutting a square of side x from each corner. A student writes $V = x(20 - x)^2$.

(a) Explain which factor of the student's volume formula is wrong and why.

Answer: _____

(b) The corrected volume satisfies $V'(x) = (20 - 2x)(20 - 6x)$. Find both critical values.

Answer: _____

(c) Say which critical value maximizes the volume, and why the other is rejected.

Answer: _____

6. A student minimizes $f(x) = x + \frac{9}{x}$ on the domain $x > 0$. They compute $f'(x) = 1 - \frac{9}{x^2}$, solve $f'(x) = 0$, and report “the minimum is at $x = -3$.”

(a) Name the error in the student’s conclusion.

Answer: _____

(b) List both solutions of $f'(x) = 0$.

Answer: _____

(c) Give the minimum value of f on $x > 0$.

Answer: _____

7. A street lamp sits atop a 15 ft pole. A person 6 ft tall walks away from the base at 4 ft/s. Let x be the person’s distance from the pole and s the length of their shadow. A student writes $\frac{6}{15} = \frac{s}{x}$ and reports that the shadow lengthens at 1.6 ft/s.

(a) Describe the error in the student’s proportion.

Answer: _____

(b) Give the corrected value of $\frac{ds}{dt}$, in feet per second.

Answer: _____

8. A closed cylindrical can must hold 1000 cm^3 . A student minimizes its surface area: “Constraint $\pi r^2 h = 1000$; objective $S = 2\pi r^2 + \pi r h$. Then $h = \frac{1000}{\pi r^2}$ and $S(r) = 2\pi r^2 + \frac{1000}{r}$, so $r \approx 4.30 \text{ cm}$.”

(a) Identify the error in the student’s objective function.

Answer: _____

(b) Write the corrected $S(r)$ as a function of r alone.

Answer: _____

(c) Find the radius that minimizes the corrected S , to two decimals.

Answer: _____